The 312 distinguished Higman–Sims parabolics

Human Compression project

This component concerns the actual distinguished parabolic subgroups in the 86 retained Higman–Sims models and the corresponding 312 node labels in the fixed supplement. The source-to-model type permutation, exact subgroup orders and actual generators are acquired from the accepted local-data and source-coordinate components. The 398 Borel/kernel occurrence maps are also acquired. The present task is the full parabolic and its extension over the *displayed* kernel. Neither a group order nor an unproved finite presentation is used as a naming certificate.

Theorem 1. *For all 312 source/model slots, the supplied literal certificates identify the actual parabolic, its distinguished kernel and quotient, with the splitting and direct-product assertions recorded in the supplied source ledger. The 61 raw source label strings and all parentheses and kernel boxes are retained. Exactly 26 slots require a correction, including five slots whose kernel correction was already acquired. The other labels are verified or made more precise. These are statements about the displayed subgroups, not a classification of all parabolic conjugacy classes or geometries.*

The finite statements in this theorem require the complete verification interface described below. This working manuscript carries no acceptance or whole-source credit before that interface and its distinct review close.

Literal transport

Lemma 1. *Let $M = \langle g_1, \dots, g_s \rangle$ be a finite permutation model whose complete right-generator Cayley graph is known. Suppose permutations $h_1, \dots, h_s$ define images y_m along the discovery words, with $y_1 = 1$, and every edge satisfies $y_{mg_i} = y_m h_i$. Then $m \mapsto y_m$ is a homomorphism. If its values are distinct, contain every generator of an actual subgroup P, and $|P| = |M|$, it is an isomorphism onto P.*

Proof. Induction on a positive word gives the claimed image of every product. Every element of a finite generated group has such a word, including inverses. Thus all products are respected. Distinctness proves injectivity; containment and the independently acquired order give equality with P. □

The enumerator closes from the identity under every generator and checks its resulting size; the expected size never truncates an open traversal. All non-discovery edges are checked. Hence a GAP isomorphism proposal is only a way of choosing the h_i. The proof uses the literal permutations and complete edges, with no group-recognition or presentation assumption.

For an extension let K be the actual listed kernel, and let M be the named quotient model. Enumerate K, verify its order, and check that every h_i normalizes it. The discovery lifts y_m must satisfy

$$y_m h_i y_{mg_i}^{-1} \in K.$$

The checked normalization makes coset multiplication well defined, so the same word argument gives a homomorphism $M \to \langle K, h_i \rangle / K$. Requiring all ky_m to be distinct proves injectivity and the exact order $|K||M|$. Every actual generator of P must occur among them. The acquired order $|P| = |K||M|$ then proves equality with P, including the actual kernel embedding. If all edges hold exactly, the y_m form a section. If the lifts also commute with every kernel generator, this is an internal direct product. A quotient certificate alone asserts neither splitting nor nonsplitting.

Complete section tests

Once the quotient map and ordered generators of M are fixed, a section must send g_i to one of $k_i h_i$, $k_i \in K$. Enumerating every ordered lift tuple and testing every Cayley edge therefore covers every section. An exact edge solution is automatically a right inverse of the quotient map and hence injective. Conversely every section supplies such a solution. The cap on the number of tuples rejects oversized tests; it never reports nonsplitting after a partial enumeration.

There are 4096 lift tuples for each of the three $|K| = 64$, $M = S_5$ cases, and all fail. Thus the three printed colons in these cases are false. The four extra central C_2 models used to identify unboxed full parabolics have respectively $4, 8, 8, 8$ lift tuples and $2, 4, 4, 4$ sections. These central subgroups are additional witnesses and are not substituted for the source's boxed kernels. A separate inner E_2^4/S_5 test has 256 tuples and 32 sections. Its purpose is a precise reassociation of one actual named model; it does not identify arbitrary groups sharing the same extension notation.

Names and parenthesized factors

The acquired named-model library gives 244 full-parabolic maps. Six further full groups are identified as $A_5 \times (5:4)$ by explicit disjoint actions; four are $C_2 \times M_{10}$ or $C_2 \times \mathrm{Aut}(A_6)$ through literal central sections. The remaining 58 are specified by their actual kernel/quotient extension certificates and fixed permutation actions. All 108 nontrivial boxed kernels have quotient certificates. There are 56 section certificates for these exact source kernels and 29 centralizing sections in total, including the four additional central witnesses. No failure of a discovery procedure is used as a nonsplitting assertion.

The elementary naming models are explicit: $\mathrm{GL}(3,2)$ on the seven nonzero binary vectors is $L_3(2)$; adjoining translations gives $\mathrm{E}_2^3 : L_3(2)$. Affine transformations of $\mathbb{F}_5$ give $5:4$. Signed coordinate matrices on $\mathbb{F}_3^2$ give $3^2 : D_8$. The matching-sign subgroup of $S_3 \times S_4$, over the normal Klein four in the second factor, is $2^2 : (3^2 : 2)$. The full product is $2^2 : (3^2 : 2^2)$. Finally $\mathbb{F}_9 = \mathbb{F}_3[u]/(u^2+1)$ has a primitive element $1+u$ of order8; Frobenius acts on it by cubing. Translations, multiplication and Frobenius therefore give $3^2 : (8:2) = A\Gamma L(1,9)$. Every use of these names has its own complete literal transport.

Nested accepted trees are not reassociated by notation alone. The two actual full models named $(2^4 : A_5) : 2$ have separately checked normal E_2^4 and S_5 complements. Both actual 576-order models needed for $2^4 : (3^2 : 2^2)$ have checked E_2^4 and $S_3 \times S_3$ complements. One of these is additionally identified as $S_4 \times S_4$. The quotient of slot $66 : P3$ has its own checked $\mathrm{E}_2^4 : S_5$ section. Model IDs here are local to their exact bound input files, never an order-based alias. For $A_5 : D_{12}$ the transported full model is $S_5 \times S_3$: $A_5 \lhd S_5$ has a transposition complement and $C_2 \times S_3 = D_{12}$. For $([2] \times A_6).2^2$, the stronger actual product $C_2 \times \mathrm{Aut}(A_6)$ supplies the normal intermediate $C_2 \times A_6$.

Source corrections

The full slot table records source label, physical page, source type, model type, the untouched raw expression and box span, and the exact certificate for every assertion. The following descriptions explain all new correction patterns; the five acquired kernel corrections remain explicitly attributed to their prior components.

- The three $[4.2^4] : S_5$ extensions do not split over the displayed kernel, by complete lift exhaustion.
- The ten quotients printed $2^3 \times L_3(2)$ are $2^3 : L_3(2)$. The latter affine group has trivial centre: commuting with translations forces a central element's linear part to be the identity, and a translation fixed by every linear map is zero.
- The quotient printed $2 \times D_{10}$ is $5:4$; an element of order4 distinguishes these groups. Quotients printed 2 in source3.1 and3.20 have order4 and are 2^2. Source3.4 type2 has quotient D_{12}, not S_3, in addition to its already corrected kernel; its corrected exact sequence makes no splitting assertion.
- The two unboxed $2 \times A_6$ labels have actual order1440, with groups $C_2 \times M_{10}$ and $\mathrm{Aut}(A_6)$ respectively. Source3.24 has full group $\mathrm{Aut}(A_6)$ with trivial centre. Even if its label means $(C_2 \times A_6).C_2$, the characteristic C_2 centre of that normal index-two subgroup is fixed by every conjugation and remains central in the whole group. Thus both conventional readings are excluded.
- Source4.24 type1 is $U_3(5) : 2$ and source4.28 type1 is $U_3(5)$, with orders252000 and126000 respectively. Literal named transport, not just the corrected type sizes, establishes these identifications.

The unparenthesized source4.12 type3 expression is read sequentially as $(K \times \mathrm{E}_2^2) : (\mathrm{E}_3^2 : \mathrm{E}_2^2)$. This is verified: its actual boxed K of order4 lies in a checked normal elementary group of order16, has an internal elementary complement there, and the normal group has a checked $S_3 \times S_3$ complement. The full group is $S_4 \times S_4$. Noncentrality of K in the whole group does not refute this intermediate direct-product assertion. The earlier working interpretation is retained as a reader correction, not a source error.

The generic dot is used only for an extension; it is not a nonsplitting claim. The exact permutation model and the maps disambiguate extensions when a conventional expression alone does not. No parabolic conjugacy, residual primitivity, geometry enumeration, automorphism-group or complete original recovery is inferred by this component.

Verification boundary

The final checker must bind the exact source/type join, actual generator lists and acquired orders, each named-model proof, and all proposal maps before rebuilding and checking every used literal edge/coset certificate. The small controls compare with independent full-multiplication or homomorphism-graph oracles, in ordinary and sanitized builds. Damaged models, maps and interfaces must reject. Development failures are retained: an omitted generator in one elementary constructor, an initial map filename collision, and a checker that initially expected an unavailable proposal. None supplies mathematical evidence. Whole-source and classification credit remain zero until the separate source-wide synthesis.